UNIT-I

1. What Is Data Mining? Explain how the evolution of database technology led to data mining.
2. Describe the steps involved in data mining when viewed as a process of knowledge discovery.
3. How is a data warehouse different from a database? How are they similar?
4. List and describe the five primitives for specifying a data mining task.
5. Describe why concept hierarchies are useful in data mining.
6. Discuss data Pre-Processing tasks indetail.
7. Explain the metrics used for data quality analysis.
8. Describe various methods for handling tuples with missing values for some attributes.
9. Why data cleaning is essential.
10. Explain Data Intergration.
11. Discuss Data Reduction techniques.
12. Explain Data Transformation Strategies.
13. For the attribute age: 13, 15, 16, 16, 19, 20,20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

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1. Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.
2. Use min-max normalization to transform the value 35 for age onto the range [0.0, 1.0].
3. Use z-score normalization to transform the value 35 for age, where the standard deviation of age is 12.94 years.
4. Use normalization by decimal scaling to transform the value 35 for age.

14. Given data: 200, 300, 400, 600, 1000. Normalize the data using the methods :
    1. min-max normalization by setting min = 0 and max = 1
    2. z-score normalization
    3. z-score normalization using the mean absolute deviation instead of standard deviation
    4. normalization by decimal scaling
15. A Survey of **2000** coffee drinkers is given below. Find the Chi-square value

Response	% of coffee drinkers	Oberserved Frequency
2 Cups Per week	20%	306
1 Cup Per week	15 %	293
2 Cups per day	27 %	562
1 Cup per day	38 %	839